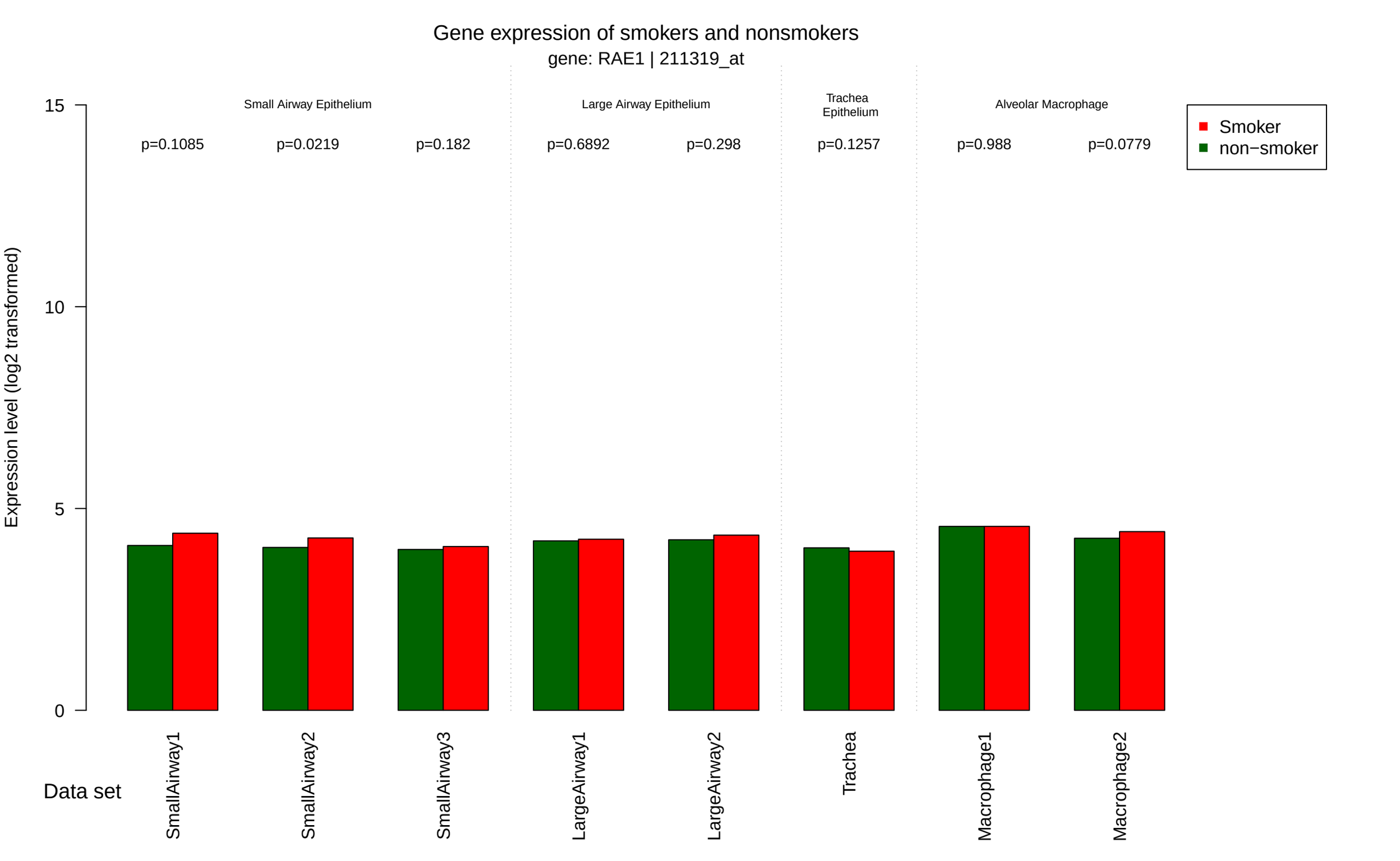
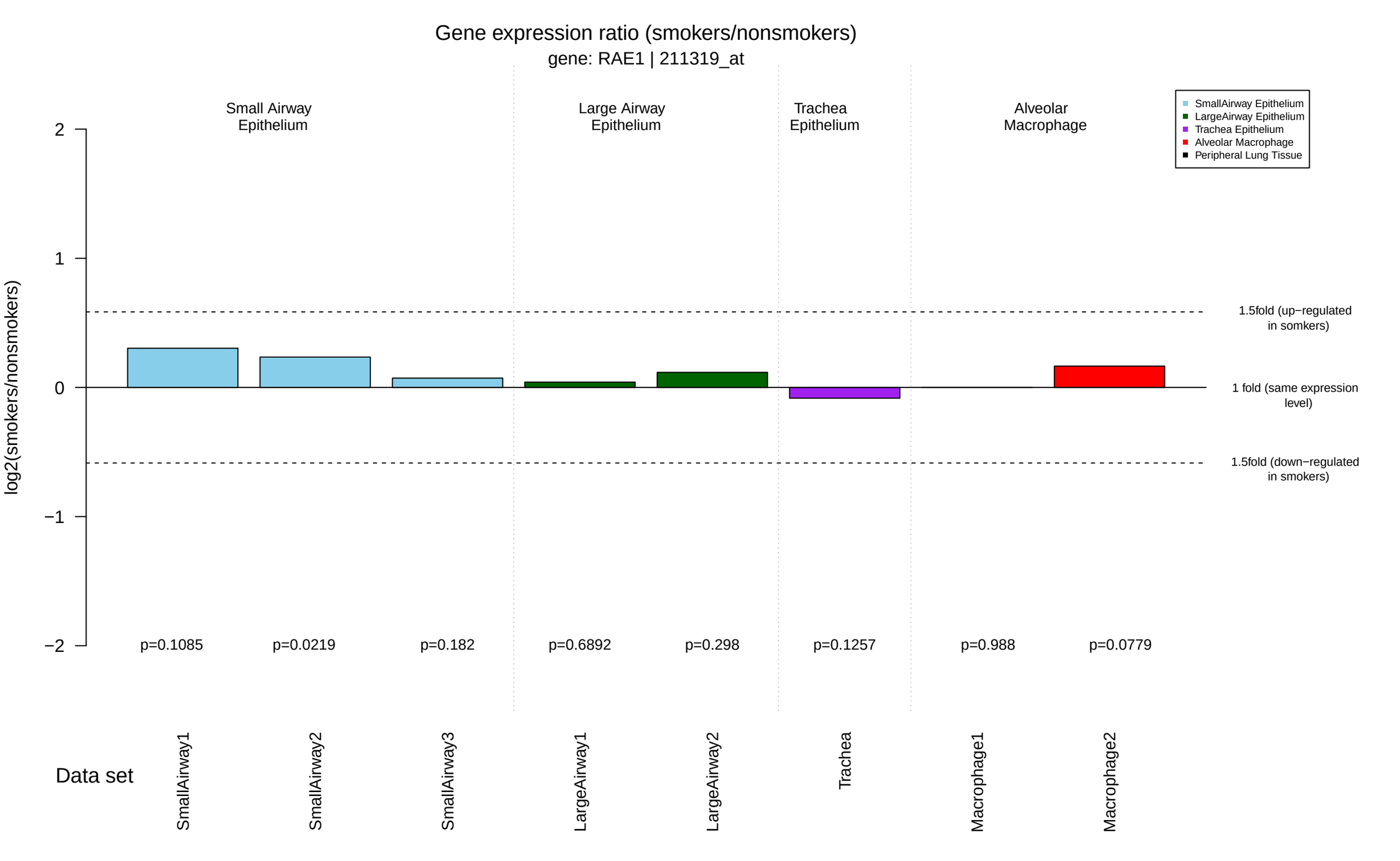
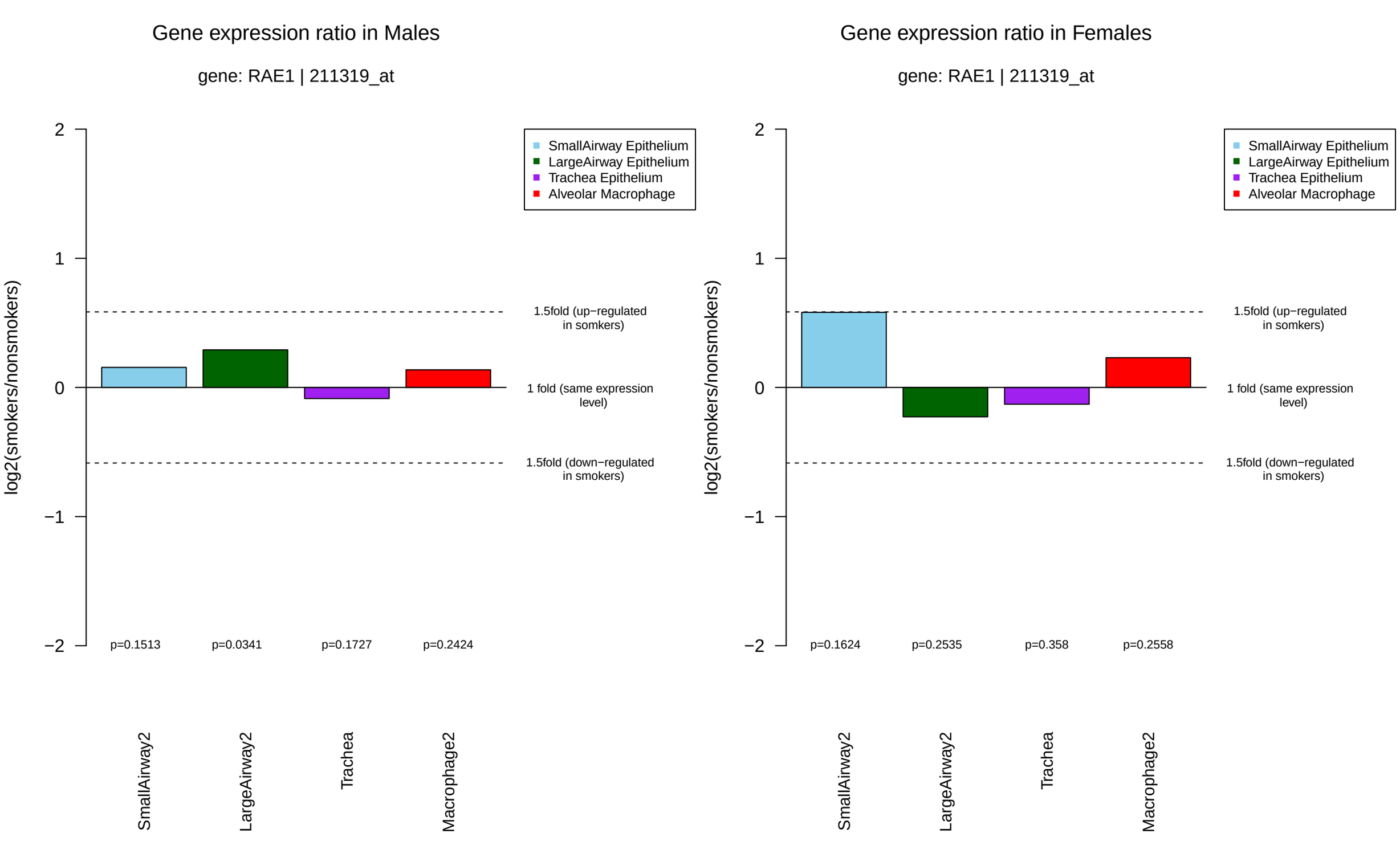
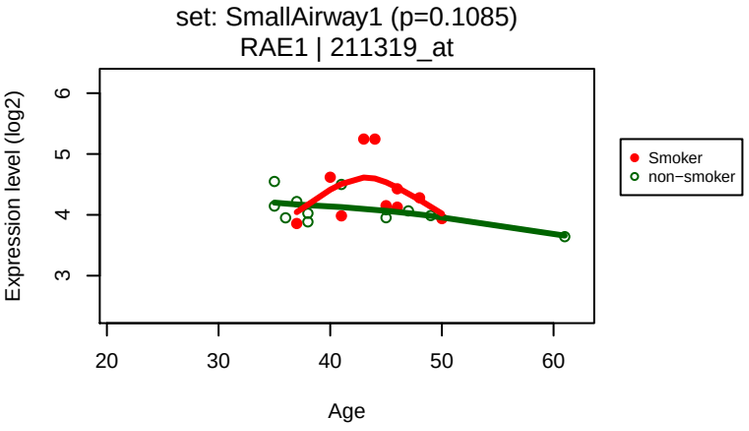
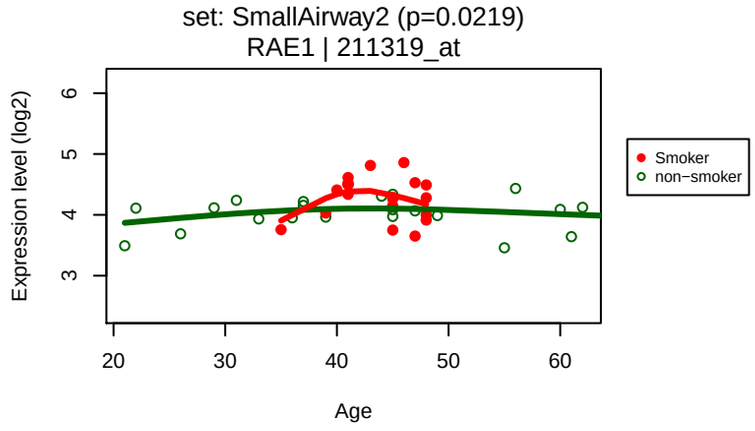
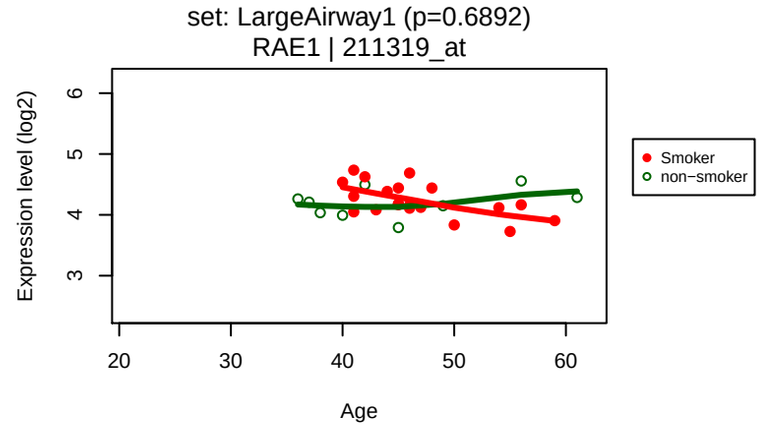

Gene expression of smokers and nonsmokers
gene: RAE1 | 211319_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
p=0.1085
p=0.0219
p=0.182
p=0.6892
p=0.298
p=0.1257
p=0.988
p=0.0779
Smoker
non−smoker
Expression level (log2 transformed)
15
10
5
0
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

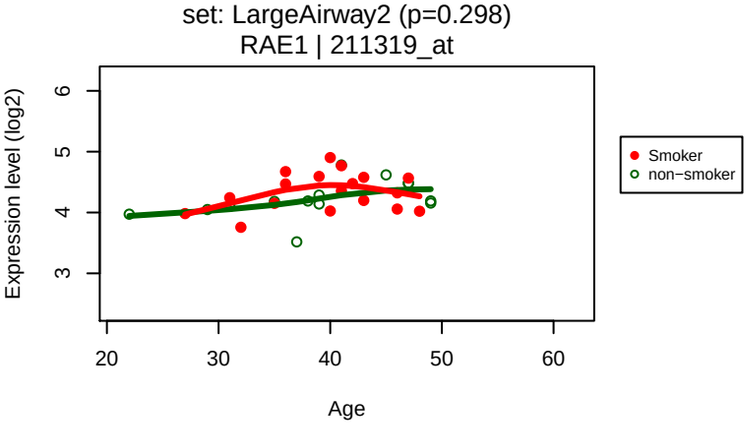

Gene expression ratio (smokers/nonsmokers)
gene: RAE1 | 211319_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Peripheral Lung Tissue
log2(smokers/nonsmokers)
2
1
0
−1
−2
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.1085
p=0.0219
p=0.182
p=0.6892
p=0.298
p=0.1257
p=0.988
p=0.0779
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

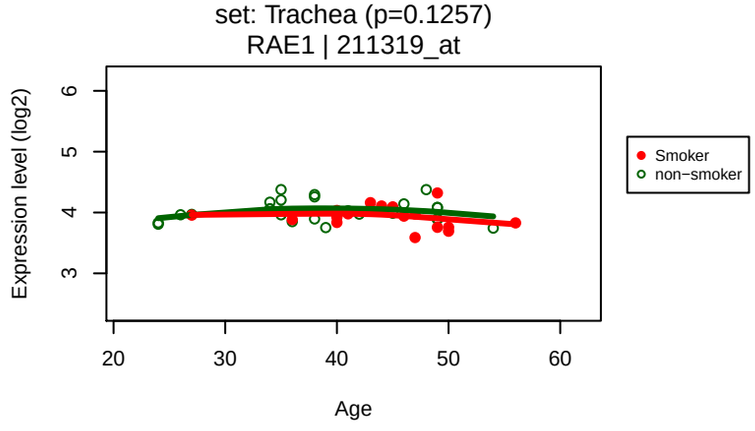

Gene expression ratio in Males
gene: RAE1 | 211319_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.1513
p=0.0341
p=0.1727
p=0.2424
SmallAirway2
LargeAirway2
Trachea
Macrophage2
Gene expression ratio in Females
gene: RAE1 | 211319_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.1624
p=0.2535
p=0.358
p=0.2558
SmallAirway2
LargeAirway2
Trachea
Macrophage2

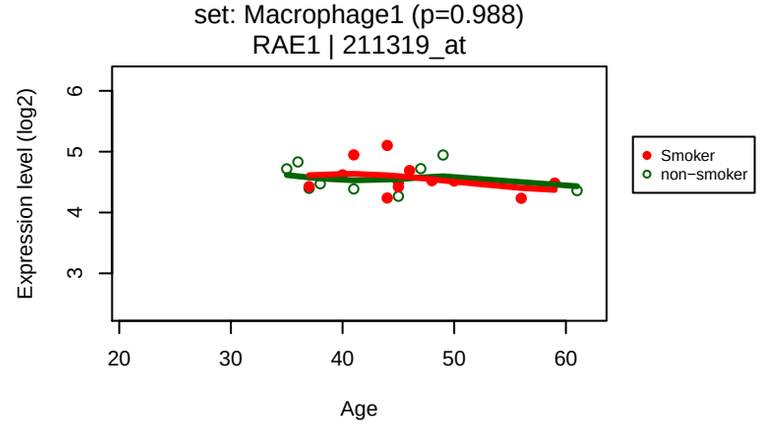

set: SmallAirway1 (p=0.1085)
RAE1 | 211319_at
Expression level (log2)
Age
Smoker
non−smoker

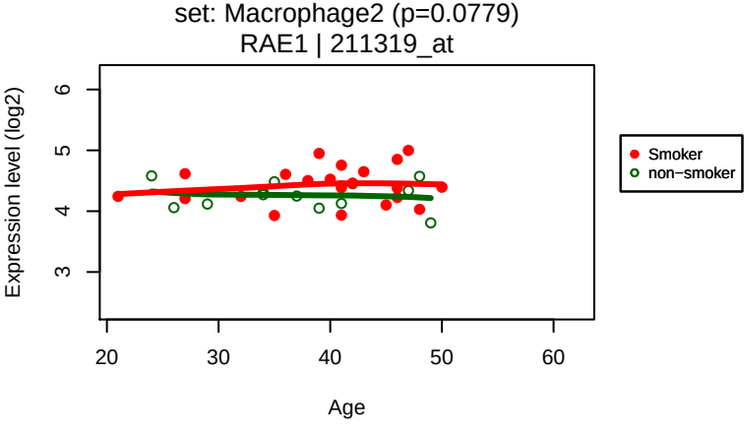

set: SmallAirway2 (p=0.0219)
RAE1 | 211319_at
Expression level (log2)
Age
Smoker
non−smoker

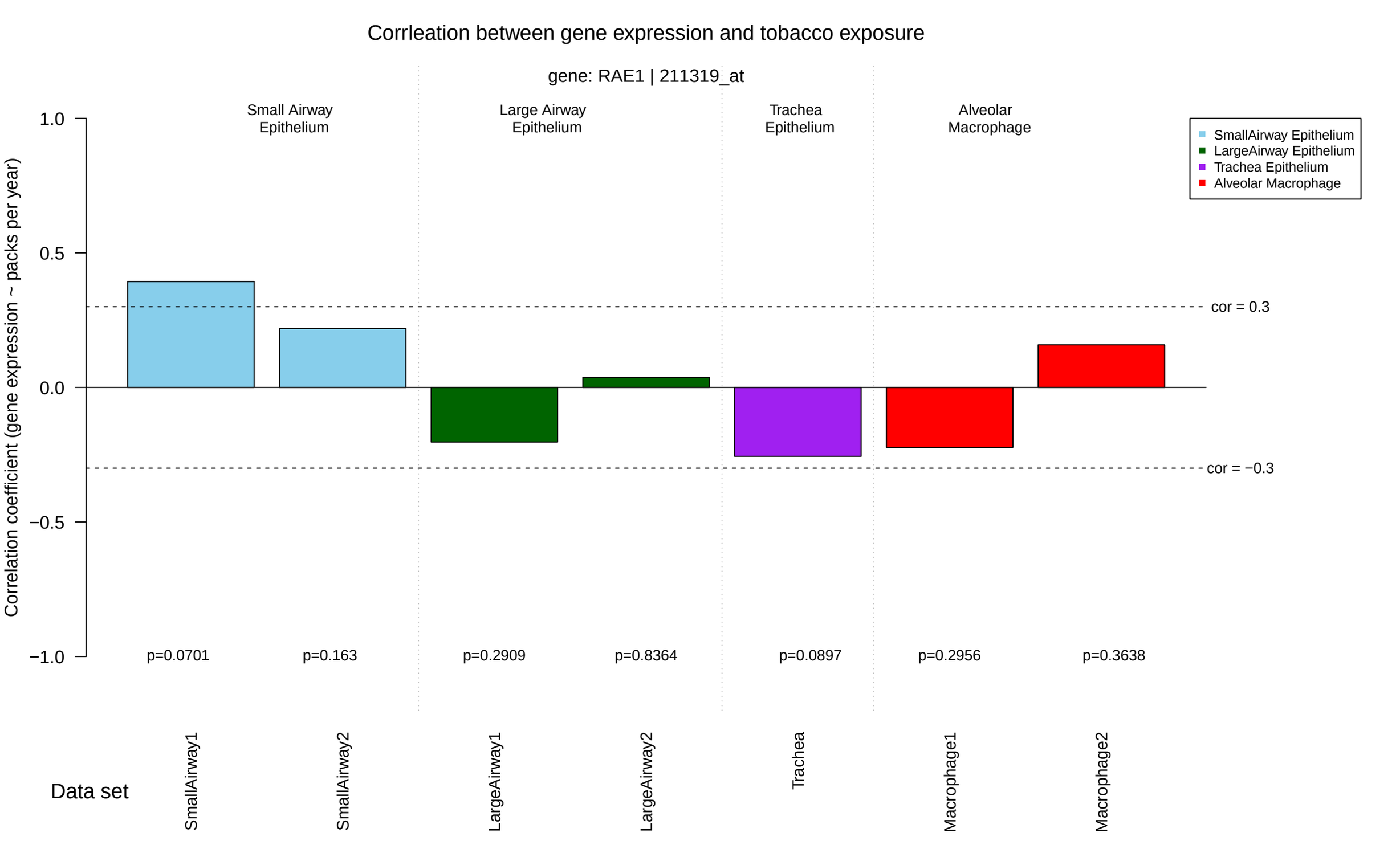

set: LargeAirway1 (p=0.6892)
RAE1 | 211319_at
Expression level (log2)
Age
Smoker
non−smoker


set: LargeAirway2 (p=0.298)
RAE1 | 211319_at
Expression level (log2)
Age
Smoker
non−smoker


set: Trachea (p=0.1257)
RAE1 | 211319_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage1 (p=0.988)
RAE1 | 211319_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage2 (p=0.0779)
RAE1 | 211319_at
Expression level (log2)
Age
Smoker
non−smoker


Corrleation between gene expression and tobacco exposure
gene: RAE1 | 211319_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Correlation coefficient (gene expression ~ packs per year)
1.0
0.5
0.0
−0.5
−1.0
cor = 0.3
cor = −0.3
p=0.0701
p=0.163
p=0.2909
p=0.8364
p=0.0897
p=0.2956
p=0.3638
Data set
SmallAirway1
SmallAirway2
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2